

- 8 (a)** Find $\int \sin \theta \cos^n \theta \, d\theta$, where $n \neq -1$. [2]

Let $I_{m,n} = \int_0^{\frac{1}{2}\pi} \sin^m \theta \cos^n \theta \, d\theta$.

- (b)** Show that, for $m \geq 2$ and $n \geq 0$,

$$I_{m,n} = \frac{m-1}{m+n} I_{m-2,n}. \quad [5]$$

- (c) By considering the binomial expansion of $\left(z + \frac{1}{z}\right)^5$, where $z = \cos \theta + i \sin \theta$, use de Moivre's theorem to show that

$$\cos^5 \theta = a \cos 5\theta + b \cos 3\theta + c \cos \theta,$$

where a , b and c are constants to be determined. [5]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

- (d) Using the results given in parts (b) and (c), find the exact value of $I_{2,5}$. [4]

[illegible]